Lecture 8

Variational approach to stochastic partial differential equations

3.4. Compactness method for solving SPDEs in the variational setting

We keep the assumptions (V.1) and (V.3), i.e.

(V.1) $\exists\, \alpha > 0$, λ, ν such that

$$2\langle A(u), u\rangle + \|B(u) \circ \sqrt{Q}\|^2_{L_2(U,H)} \leq -\alpha\|u\|^2_V + \lambda\|u\|^2_H + \nu$$

(V.3) $\exists M$ such that

$$\|A(u)\|_{V'} \leq M(1 + \|u\|_V).$$

This time we assume in addition

(V.5) $\exists M, \delta > 0$, such that

$$\|B(u) \circ \sqrt{Q}\|_{L_2(U,H)} \leq M(1 + \|u\|_V^{1-\delta})$$

corresponding to sublinear growth since $\delta > 0$.

(V.6) $V \hookrightarrow H$ ist compact, i.e., bounded subsets of V are precompact in H.

(V.7) $u \mapsto A(u) : V_{\text{weak}} \cap H \longrightarrow V'_{\text{weak}}$ is continuous,
$u \mapsto B(u) : V_{\text{weak}} \cap H \longrightarrow L_2(U, H)$ is continuous.

Definition 3.12. A probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ is called a solution of the martingale problem associated with the SPDE (3.7)

$$du(t) = A(u(t))dt + B(u(t))dW(t), \quad u(0) = u_0$$

if

(i) $\mathbb{P}(u(0) = u_0) = 1$,

(ii) the stochastic process

$$M_t := u(t) - u(0) - \int_0^t A(u(s))ds$$

is a continuous H-valued $\mathbb{P}$-martingale with associated increasing process

$$\langle M\rangle_t = \int_0^t B(u(s)) \circ Q \circ B(u(s))^* ds.$$

The second condition in the above definition is equivalent to the following condition:

(ii') Let (e_n) be a CONS of H consisting of elements $e_k \in V$. Then for all $i \geq 1$, $\varphi \in C_b^2(\mathbb{R})$, $0 \leq s \leq t$ and any Φ_s continuous, bounded and $\mathcal{F}_s$-measurable:

$$\mathbb{E}_n\left[(M_t^{i,\varphi} - M_s^{i,\varphi})\Phi_s\right] = 0,$$

where

$$M_t^{i,\varphi} := \varphi(\langle u(t), e_i\rangle_H) - \varphi(\langle u_0, e_i\rangle) - \left(\int_0^t \varphi'(\langle u(s), e_i\rangle)\langle A(u(s)), e_i\rangle ds + \frac{1}{2}\int_0^t \varphi''(\langle u(s), e_i\rangle\langle (B\circ Q\circ B^*)(u(s))e_i, e_i\rangle ds \right).$$

THEOREM 3.13. *Under the assumptions (V.1), (V.3) and (V.5)-(V.7) there exist a solution $\mathbb{P}$ to the martingale problem associated with* (3.7).

PROOF. We consider the same Galerkin approximation as in the proof of 3.5: $V_n := \text{span}\{e_1, ..., e_n\}$ for some CONS (e_n) of H consisting of elements $e_k \in V$. Let Π_n be the orthogonal projection in H onto V_n. Then there exists for all $n \geq 1$ a solution $\mathbb{P}_n$ on $(\Omega, \mathcal{F})$, such that

$(0)_n$ $\text{supp}(\mathbb{P}_n) \subset C([0,T]; V_n)$
$(i)_n$ $\mathbb{P}_n(u(0) = \Pi_n u_0) = 1$
$(ii)_n$ $\forall \varphi \in C_b^2(\mathbb{R})$ and Φ_s continuous, bounded and $\mathcal{F}_s$-measurable

$$\mathbb{E}_n\left[(M_t^{i,\varphi} - M_s^{i,\varphi})\Phi_s\right] = 0.$$

$\mathbb{P}_n$ is obtained as the law of the corresponding finite-dimensional stochastic differential equation.

Suppose now the following lemma holds:

LEMMA 3.14. *The sequence of probability measures $(\mathbb{P}_n)_{n\in\mathbb{N}}$ is tight on*

$$\Omega := C([0,T]; V'_{weak}) \cap L^2([0,T]; V) \cap L^2([0,T]; H)$$

w.r.t. the three topologies τ_i, $i = 1, 2, 3$, where

- *τ_1 is the weak topology on $L^2([0,T]; V)$,*
- *τ_2 is the uniform topology on $C([0,T]; V'_{weak})$,*
- *τ_3 is the strong topology on $L^2([0,T]; H)$.*

Continuing the proof of Theorem 3.13

Lemma 3.14 implies that we can extract a subsequence, again denoted with $(\mathbb{P}_n)_n$, such that $\mathbb{P}_n \overset{w}{\Rightarrow} \mathbb{P}$ weakly on $C([0,T]; V) \subset \underbrace{C([0,T]; H)}_{=\Omega}$. The limiting probability measure $\mathbb{P}$ satisfies condition (i) and the mapping

$$\Omega \longrightarrow \mathbb{R}, \quad \omega \mapsto (M_t^{i,\varphi}(\omega) - M_s^{i,\varphi}(\omega))\Phi_s(\omega)$$

is continuous on Ω.

The coercivity assumption (V.1) together with Lemma 3.9 implies that

$$\sup_{n\in\mathbb{N}} \mathbb{E}_n \left(\sup_{t\in[0,T]} \|u(t)\|^2 + \int_0^T \|u(t)\|_V^2 ds \right) < \infty.$$

In particular, we obtain for some $p > 1$ (depending on δ in (V.5))

$$\sup_{n\in\mathbb{N}} \mathbb{E}_n \left(|M_t^{i,\varphi} - M_s^{i,\varphi}|^p \right) < \infty,$$

since,

$$\begin{aligned} M_t^{i,\varphi} - M_s^{i,\varphi} = & \underbrace{\varphi(\langle u(t), e_i\rangle) - \varphi(\langle u(s), e_i\rangle)}_{\text{bounded}} \\ & + \int_s^t \underbrace{\varphi'(\langle u(r), e_i\rangle)}_{\text{bounded}} \underbrace{\langle A(u(r)), e_i\rangle}_{\overset{\text{(V.3)}}{\leq} M(1+\|u(r)\|_V)\|e_i\|_V} dr \\ & + \frac{1}{2}\int_s^t \underbrace{\varphi''(\langle u(r), e_i\rangle)}_{\text{bounded}} \underbrace{\langle (B\circ Q\circ B^*)(u(r))e_i, e_i\rangle}_{\overset{\text{(V.5)}}{\leq} M\left(1+\|u(r)\|_V^{2(1-\delta)}\|e_i\|_V^2\right)} dr, \end{aligned}$$

so that

$$\left| M_t^{i,\varphi} - M_s^{i,\varphi} \right| \underset{\delta\leq\frac{1}{2}}{\overset{\text{w.l.o.g.}}{\leq}} M_{s,t,\varphi,\|e_i\|_V,T} \left(1 + \int_0^T \|u(s)\|_V^2 ds \right)^{1-\delta}$$

and we can choose $p = \frac{1}{1-\delta} > 1$.

It follows that

$$\lim_{n\to\infty} \mathbb{E}_n \left((M_t^{i,\varphi} - M_s^{i,\varphi})\Phi_s \right) = \mathbb{E}\left((M_t^{i,\varphi} - M_s^{i,\varphi})\Phi_s \right) = 0,$$

which implies (ii). Indeed, for all $K > 0$,

$$\Psi_K := (-K) \vee \left(M_t^{i,\varphi} - M_s^{i,\varphi} \right) \wedge K$$

is bounded and continuous. Furthermore,

$$\begin{aligned} & \left| \mathbb{E}_n[(\Psi_K - (M_t^{i,\varphi} - M_s^{i,\varphi}))\Phi_s] \right| \\ & \qquad \leq \|\Phi_s\|_\infty \mathbb{E}_n \left(|M_t^{i,\varphi} - M_s^{i,\varphi}| 1_{\{|M_t^{i,\varphi} - M_s^{i,\varphi}| \geq K\}} \right) \\ & \qquad \leq \|\Phi_s\|_\infty \frac{1}{K^{p-1}} \mathbb{E}_n \left(|M_t^{i,\varphi} - M_s^{i,\varphi}|^p \right) \\ & \qquad \underset{\text{for } K\nearrow\infty}{\overset{\text{unif. in } n}{\longrightarrow}} 0. \end{aligned}$$

Consequently for all K:

$$\begin{aligned} & \lim_{n\to\infty} \left| \mathbb{E}_n[(M_t^{i,\varphi} - M_s^{i,\varphi})\Phi_s] - \mathbb{E}[(M_t^{i,\varphi} - M_s^{i,\varphi})\Phi_s] \right| \\ & \quad \leq \|\Phi_s\|_\infty \limsup_{n\to\infty} \left(\mathbb{E}_n(|(M_t^{i,\varphi} - M_s^{i,\varphi}) - \Psi_K|) + \mathbb{E}(|(M_t^{i,\varphi} - M_s^{i,\varphi}) - \Psi_K|)) \right) \\ & \quad \overset{K\nearrow\infty}{\longrightarrow} 0. \end{aligned}$$

□

It remains to prove Lemma 3.14.

Proof of Lemma 3.14. Let

- τ_1 be the weak topology on $L^2([0,T];V)$,
- τ_2 be the uniform topology on $C([0,T];V'_{weak})$,
- τ_3 be the strong topology on $L^2([0,T];H)$.

We will show that $(\mathbb{P}_n)_n$ is tight w.r.t. all three topologies. It will then follow that there exists some subsequence, again denoted with $(\mathbb{P}_n)_n$, and a probability measure $\mathbb{P}$ on $C([0,T];V') \cap L^2([0,T];V)$ such that

$$\mathbb{E}_n(\Phi(u)) = \lim_{n\to\infty} \int \Phi \, d\mathbb{P}_n = \int \Phi \, d\mathbb{P}$$

for all bounded Φ that are continuous w.r.t. τ_i, for some $i = 1,2,3$.

For given l and k let

$$K := \{u \in C([0,T];V') \cap L^2([0,T];V) \mid u(0) = u_0, \sup_{t\in[0,T]} \|u(t)\|_H \le l, \int_0^T \|u(t)\|_V^2 dt \le k\}.$$

The a priori moment estimates on the Galerkin approximation imply that $\mathbb{P}_n(K^c)$ can be made arbitrarily small for large l and k uniformly in $n \in \mathbb{N}$.

1. τ_1-tightness Consider the set

$$K_1 := \{u \in L^2([0,T];V) \mid \int_0^T \|u(t)\|_V^2 dt \le k\}.$$

Then K_1 is relatively weakly compact (Banach-Alaoglu) and

$$\sup_n \mathbb{P}_n(K_1) \le \frac{1}{k}\mathbb{E}_n\left(\int_0^T \|u(t)\|_V^2 \, dt\right) \to 0, k \to \infty.$$

2. τ_2-tightness This time we consider the set

$$K_2 := \{u \in C([0,T];V'_{weak}) \mid \sup_{0\le t\le T} \|u(t)\|_H \le k, \sup_{0\le s<t\le T} \frac{\|u(t)-u(s)\|_{V'}}{|t-s|^\beta} \le k\}$$

for some $\beta > 0$. Then K_2 is relatively compact by Arzela-Ascoli's theorem (combined with Banach-Alaoglu), since

(i) $\sup_{u\in K_2} \|u(t)\|_{V'} \le C \sup_{u\in K_2} \|u(t)\|_H < \infty$
(ii) $\sup_{u\in K_2} \sup_{|t-s|\le\delta} \|u(t)-u(s)\|_{V'} \le k\delta^\alpha \longrightarrow 0$ for $\delta \searrow 0$.

We then have $\mathbb{P}_n(K_2) \to 0$ as $k \to \infty$, since w.r.t. $\mathbb{P}_n$ for $\|h\|_V \le 1$

$$\begin{aligned}|\langle u(t),h\rangle - \langle u(s),h\rangle| &= \left|\int_s^t \langle A(u(r),h\rangle dr + \int_s^t \langle h, B^n(u(r))dW^n(r)\rangle\right| \\ &\le M\left(|t-s| + \sqrt{|t-s|}\int_s^t \|u(r)\|_V^2 dr\right) + \underbrace{\left|\int_s^t \langle h, B^n(u(r))dW^n(r)\rangle\right|}_{(*)}\end{aligned}$$

and

$$
\begin{aligned}
&\mathbb{E}_n\left(\sup_{r\in[s,t]}\left|\int_s^r \langle h, B^n(u(r))dW^n(r)\rangle\right|^p\right)\\
&\qquad = \mathbb{E}_n\left(\sup_{r\in[s,t]}\left|\langle h, \int_s^r B^n(u(r))dW^n(r)\rangle\right|^p\right)\\
&\overset{\substack{\text{Burkholder-}\\\text{Davis-}\\\text{Gundy}}}{\leq} c_p\mathbb{E}_n\left(\left(\int_s^t \|h\|_H^2 \underbrace{\|B(u(r))\circ\sqrt{Q}\|^2_{L_2(U,H)}}_{\leq M(1+\|u(r)\|_V^{1-\delta})^2} dr\right)^{\frac{p}{2}}\right)\\
&\qquad \leq 2^{\frac{p}{2}}c_pM^p \underbrace{\|h\|_H^p}_{\leq 1}\mathbb{E}_n\left(\left((t-s)+\int_s^t \|u(r)\|_V^{2(1-\delta)}\,dr\right)^{\frac{p}{2}}\right)\\
&\qquad \leq 2^{\frac{p}{2}+1}c_pM^p\left((t-s)^{\frac{p}{2}}+(t-s)^{\frac{p}{2}\delta}\mathbb{E}_n\left(\int_s^t \|u(r)\|_V^2\,dr\right)^{\frac{p}{2}(1-\delta)}\right)\\
&\qquad \leq 2^{\frac{p}{2}+1}c_pM^p\left((t-s)^{\frac{p}{2}}+\sup_n\left(\mathbb{E}_n\left(\int_s^t \|u(r)\|_V^2\,dr\right)\right)^{\frac{2}{p(1-\delta)}}(t-s)^{\frac{p}{2}\delta}\right)
\end{aligned}
$$

so that

$$
\sup_n \mathbb{P}_n(K_2) \leq \frac{C}{k}\mathbb{E}_n\left(\sup_{0\leq t\leq T}\|u(t)\|_H^2+\int_0^T \|u(t)\|_V^2\,dt\right)\to 0, k\to\infty.
$$

3. τ_3-tightness This will follow from the next lemma. □

LEMMA 3.15. *Given a sequence $(u_n)_n$, bounded in $L^2([0,T];V)\cap L^\infty([0,T];H)$ and equicontinuous as V'-valued functions, such that $u_n(0)\longrightarrow u_0$ strongly in H, one can extract a subsequence strongly convergent in $L^2([0,T];H)$.*

PROOF. We will use the following fact: $\forall\varepsilon>0\ \exists c_\varepsilon$ such that for all $u\in V$:

$$
\|u\|_H \leq \varepsilon\|u\|_V + c_\varepsilon\|u\|_{V'}.
$$

Indeed, otherwise we could find $\varepsilon>0$ and a sequence $(u_n)_n$ such that

$$
\|u_n\|_H \geq \varepsilon\|u_n\|_V + n\|u_n\|_{V'}.
$$

But then, using $\tilde{u}_n := \frac{u_n}{\|u_n\|_H}$, and

$$
1 = \|\tilde{u}_n\|_H \geq \varepsilon\|\tilde{u}_n\|_V + n\|\tilde{u}_n\|_{V'},
$$

we can extract a weakly convergent subsequence $\tilde{u}_n\to\tilde{u}\in V$ for some $\tilde{u}\in V$, hence strongly convergent in H. Since

$$
\|\tilde{u}\|_{V'} = \lim_{n\to\infty}\|\tilde{u}_n\|_{V'} \leq \frac{1}{n}\overset{n\to\infty}{\longrightarrow} 0,
$$

we obtain that $\tilde{u}=0$, which gives a contradiction, since $\tilde{u}\neq 0$.

We know that there exists a subsequence $(u_n)_n$ converging to some $u\in C([0,T];V')$, hence also in $L^2([0,T];V')$. Clearly, $u\in L^2([0,T];V)$, hence for all $\varepsilon>0$ there

exists some c_ε such that

$$\begin{aligned}\limsup_{n\to\infty}\int_0^T \|u_n(t)-u(t)\|_H^2\,dt &\le \limsup_{n\to\infty}\varepsilon\int_0^T \|u_n(t)-u(t)\|_V^2\,dt \\ &\quad \underbrace{+\limsup_{n\to\infty} c_\varepsilon\int_0^T \|u_n(t)-u(t)\|_{V'}^2\,dt}_{=0} \\ &\le \varepsilon T \sup_{t\in[0,T]}\|u_n(t)-u(t)\|_V^2.\end{aligned}$$

□